

Risk Management, Firewall, DMZ, and Firewall Implementation

1. Risk Management

Definition

Risk management is the process of:

- Identifying risks
- Analyzing risks
- Reducing or controlling risks

in a computer network or organization.

Basic Terms

Term	Meaning
Asset	Valuable resource (server, data, PC)
Threat	Something harmful (virus, hacker)
Vulnerability	Weakness in system
Risk	Chance of damage due to threat exploiting vulnerability
Countermeasure	Protection method

Risk Management Process

Identify Risk	Analyze Risk	Apply Security	Monitor Risk
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Real Life Example

Bank Example

A bank stores customer money and data online.

Risks

- Hackers stealing money
- Server crash
- Employee leaking data

Exposure

- Weak passwords
- No firewall
- Outdated software

Countermeasures

- Strong passwords
 - Firewall installation
 - Antivirus
 - Data backup
 - Employee training
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Another Real Life Example

College Wi-Fi Network

Risk

Students may hack the attendance server.

Exposure

Default admin password is enabled.

Countermeasure

Change default password and enable firewall rules.

2. Firewall

Definition

A firewall is a security system that monitors and controls incoming and outgoing network traffic.

It acts like a security guard between:

- Trusted network
 - Untrusted network (Internet)
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Real Life Example

Security Guard Example

Firewall works like a guard at a company gate.

- Authorized people Allowed
- Unauthorized people Blocked

Similarly:

- Safe traffic Allowed
 - Malicious traffic Blocked
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Types of Firewalls

Type	Description
Packet Filtering Firewall	Filters packets using IP and ports
Proxy Firewall	Checks application-level traffic
Stateful Firewall	Tracks active connections

Example

Home Router Firewall

Your Wi-Fi router blocks unknown incoming traffic automatically.

That is a firewall working in real life.

3. De-Militarized Zone (DMZ)

Definition

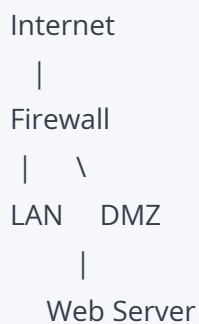
A DMZ is a separate network zone used to host public services securely.

Examples:

- Web server
- Mail server
- DNS server

DMZ protects the internal network from direct internet access.

DMZ Architecture



Real Life Example

Shopping Mall Example

Think of:

- Mall entrance = DMZ
- Staff office = Internal network

Visitors can enter the mall area,
but cannot directly enter staff offices.

Similarly:

- Users can access web server in DMZ
 - Attackers cannot directly access internal LAN
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Why DMZ is Important

Benefit	Description
Better Security	Protects internal network
Controlled Access	Public users access only DMZ
Isolation	Server attacks stay isolated

Real Example

Companies place:

- Website servers
- Email servers

inside DMZ so attackers cannot directly reach employee systems.

4. Two Methods of Implementing Firewall

There are mainly two common methods:

Method 1: Packet Filtering Firewall

Definition

Checks packets based on:

- IP address
- Port number
- Protocol

using rules.

Working

Example rule:

```
Allow TCP port 80
Block ICMP traffic
```

Real Life Example

Apartment Gate Example

Security guard checks:

- Flat number
- Visitor ID

without checking luggage contents.

Similarly:

Packet filtering checks only packet headers.

Advantages

- Fast
 - Simple
 - Low resource usage
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Disadvantages

- Cannot inspect application content
 - Less secure than proxy firewall
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Example Software

- Linux `iptables`
 - Router ACLs
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Method 2: Proxy Firewall

Definition

A proxy firewall receives traffic first, checks it completely, then forwards it to destination.

It works at application layer.

Real Life Example

Reception Desk Example

Visitor cannot directly enter office.

First:

- Reception checks identity
- Approves request
- Then allows entry

Similarly:
Proxy firewall checks web requests before forwarding.

Advantages

- High security
 - Content filtering
 - User authentication
 - Web caching
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Disadvantages

- Slower than packet filtering
 - More resource usage
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Example Software

- Squid
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Comparison Table

Feature	Packet Filtering	Proxy Firewall
Layer	Network Layer	Application Layer
Speed	Fast	Slower
Security	Basic	Advanced
Content Inspection	No	Yes
Example	iptables	Squid
